Acute Kidney Injury and Chronic Kidney Disease



AGENDA

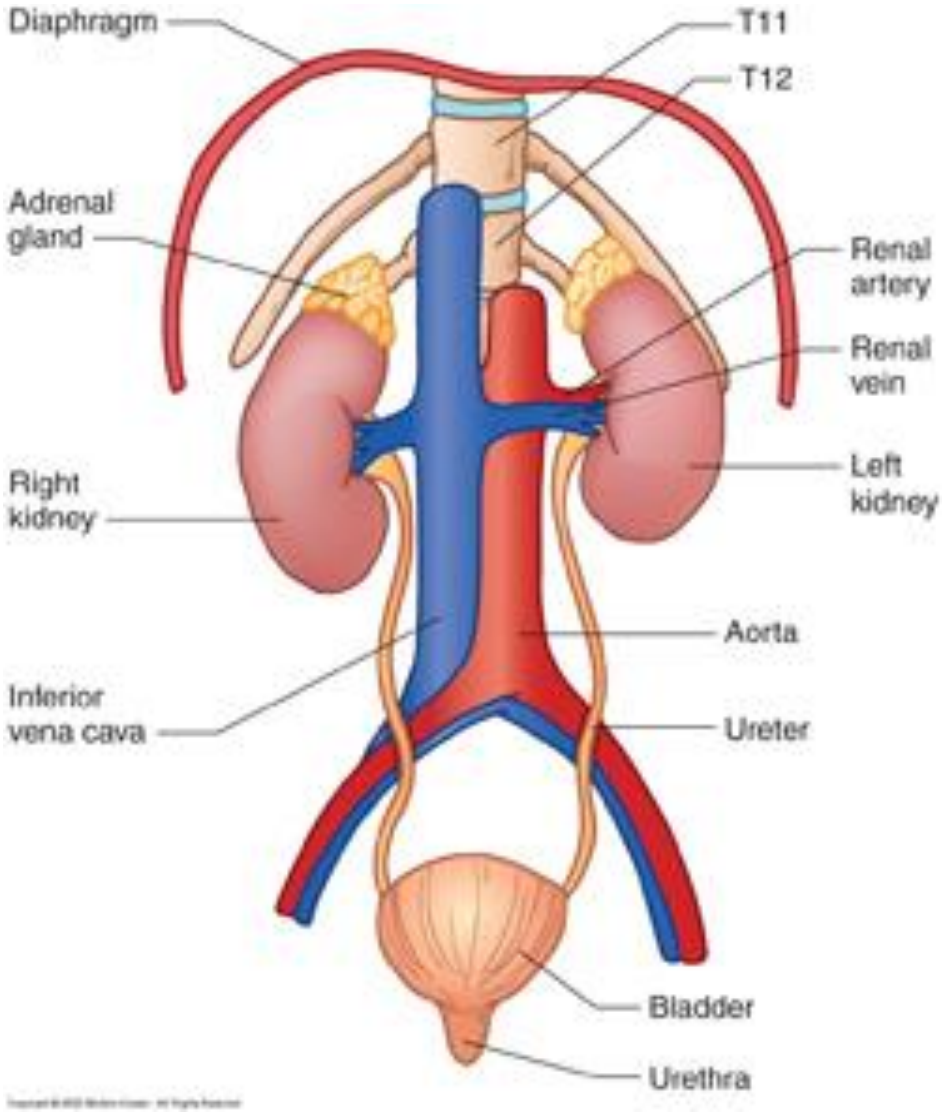
1. Explain the pathophysiology of Acute Kidney Injury (AKI) and Chronic Kidney Disease (CKD), including their causes, progression, and impact on renal function.
2. Differentiate between AKI and CKD based on clinical presentation, diagnostic criteria, and management approaches.
3. Describe the principles of renal replacement therapy (dialysis and kidney transplantation), including indications, types, and nursing considerations for patient care.



- https://youtu.be/utinirNuXeQ?si=1by__c4V7mvmF0lO

OVERVIEW OF KIDNEY FUNCTION

- Filtration of blood
- Waste removal (urea, creatinine)
- Regulation of fluid, electrolytes, and acid-base balance
- Endocrine functions (EPO, vitamin D activation, RAAS)

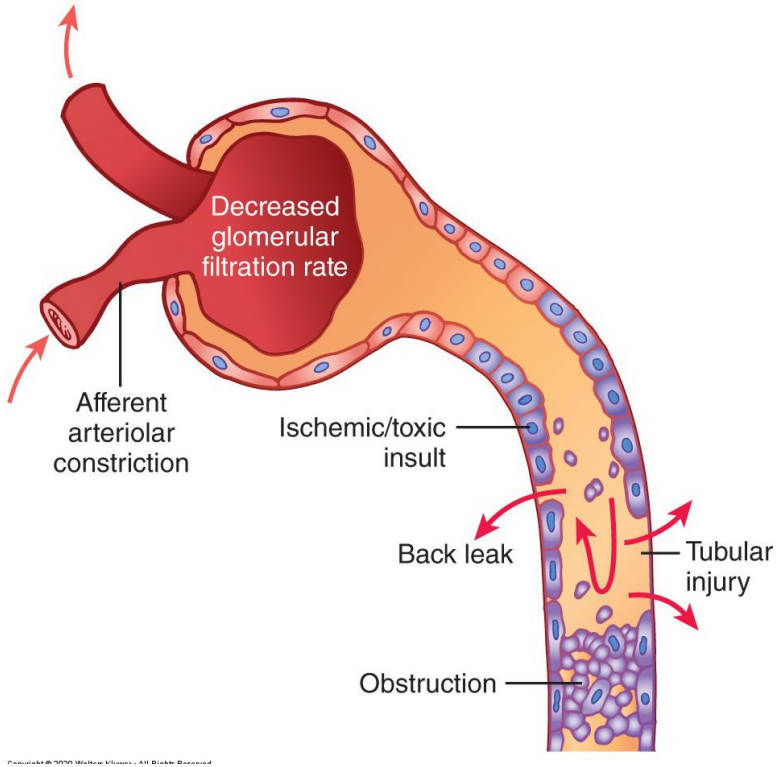


ACUTE KIDNEY INJURY (AKI)

- **Definition of AKI**
- Sudden decline in kidney function over hours to days
- Diagnostic criteria (KDIGO, RIFLE, AKIN)
- **Causes of AKI**
- **Prerenal:** Hypovolemia, CHF, shock
- **Intrinsic:** Acute tubular necrosis (ATN), glomerulonephritis, toxins
- **Postrenal:** Obstruction (stones, tumors, BPH

Pathophysiology of AKI

- Hypoperfusion leading to ischemic injury
- Inflammation and tubular damage
- Reversible vs. irreversible changes
- **Clinical Features of AKI**
- Oliguria (<400 mL/day)
- Azotemia (↑ BUN, ↑ creatinine)
- Fluid overload, hyperkalemia, metabolic acidosis



MANAGEMENT OF AKI

- Fluid resuscitation (prerenal)
- Remove nephrotoxins
- Dialysis in severe cases

CHRONIC KIDNEY DISEASE (CKD) & CHRONIC RENAL FAILURE (CRF)

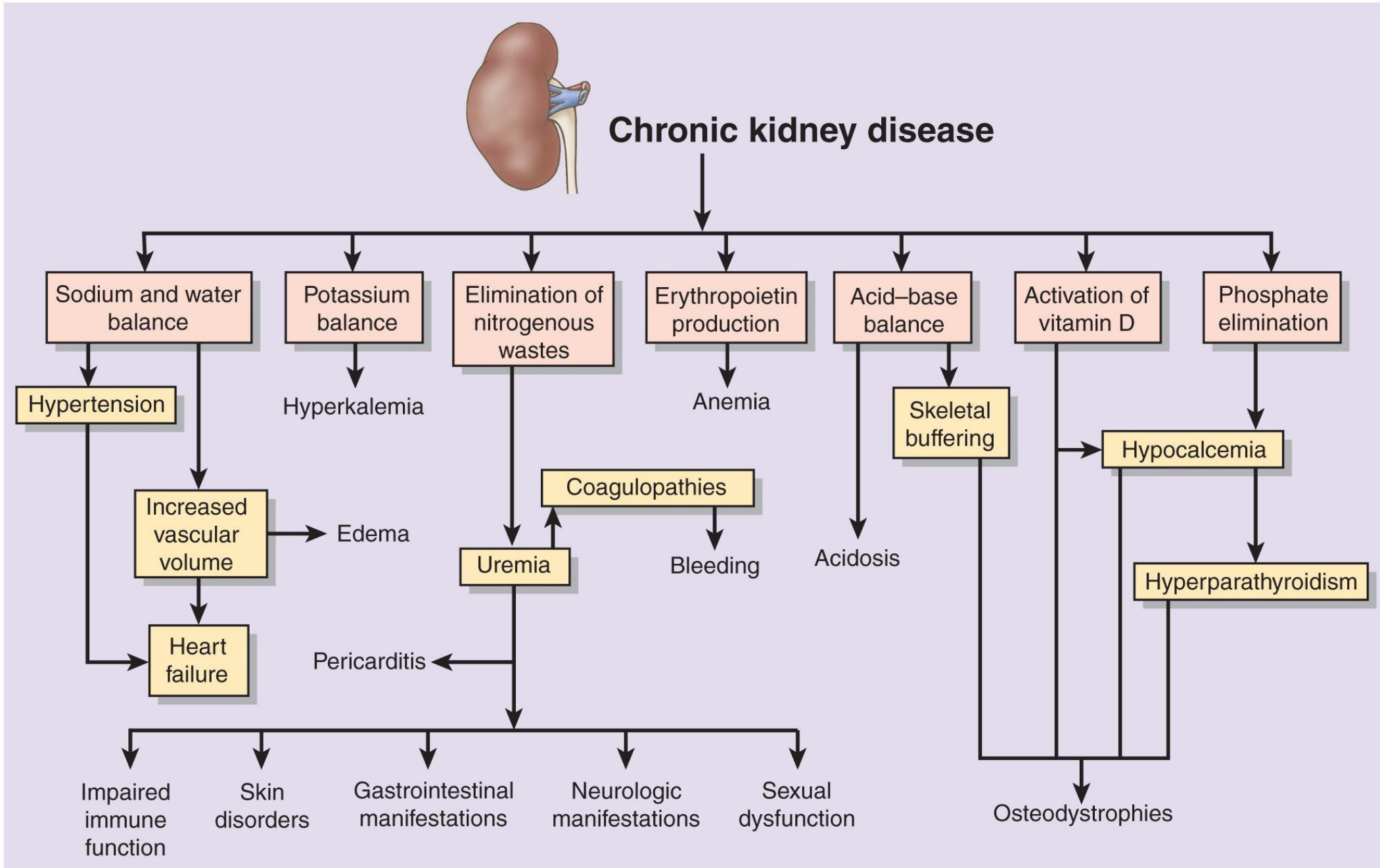
- **Definition and Stages of CKD**
- Gradual loss of kidney function over months/years
- Stages based on GFR (Normal >90, Stage 5 <15 mL/min)
- **Causes of CKD**
- Diabetes mellitus (leading cause)
- Hypertension
- Glomerulonephritis
- Polycystic kidney disease

PATHOPHYSIOLOGY OF CKD

- Nephron loss → compensatory hypertrophy
- Activation of RAAS → hypertension
- Chronic inflammation → fibrosis & scarring
- **Clinical Manifestations of CKD**
- **Early:** Asymptomatic, proteinuria
- **Late-stage:** Uremia, metabolic acidosis, hyperkalemia
- Cardiovascular complications (HTN, LVH)

COMPLICATIONS OF CHRONIC RENAL FAILURE

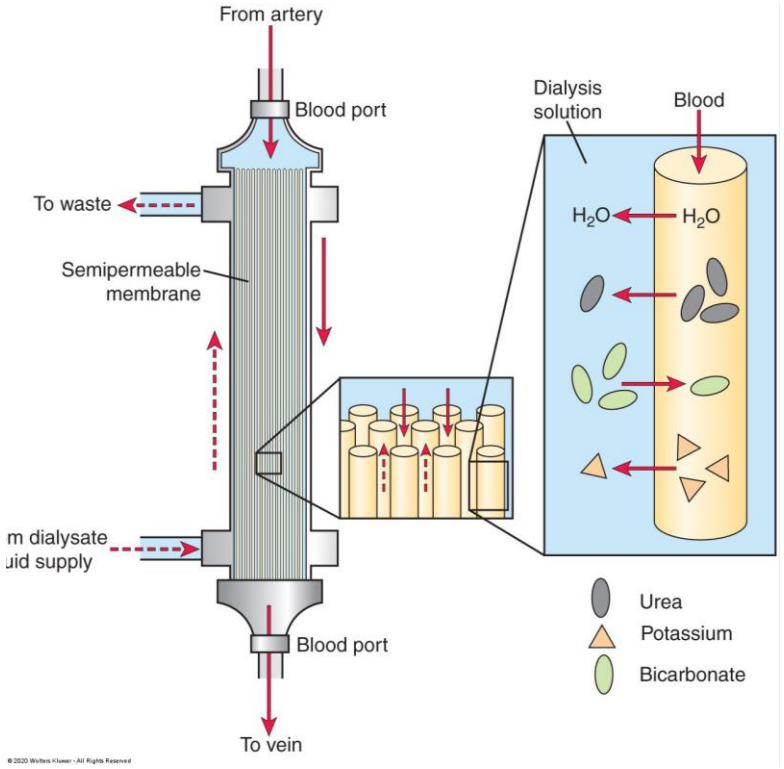
- **Cardiovascular disease**: Leading cause of death
- **Electrolyte imbalances**: Hyperkalemia, hyponatremia
- **Bone-mineral disorders**: Secondary hyperparathyroidism
- **Anemia**: Erythropoietin deficiency
- **Management of CKD**
- **Lifestyle modifications**: Low-protein diet, BP control
- **Medications**: ACE inhibitors, phosphate binders, EPO therapy
- **Dialysis and Transplantation**



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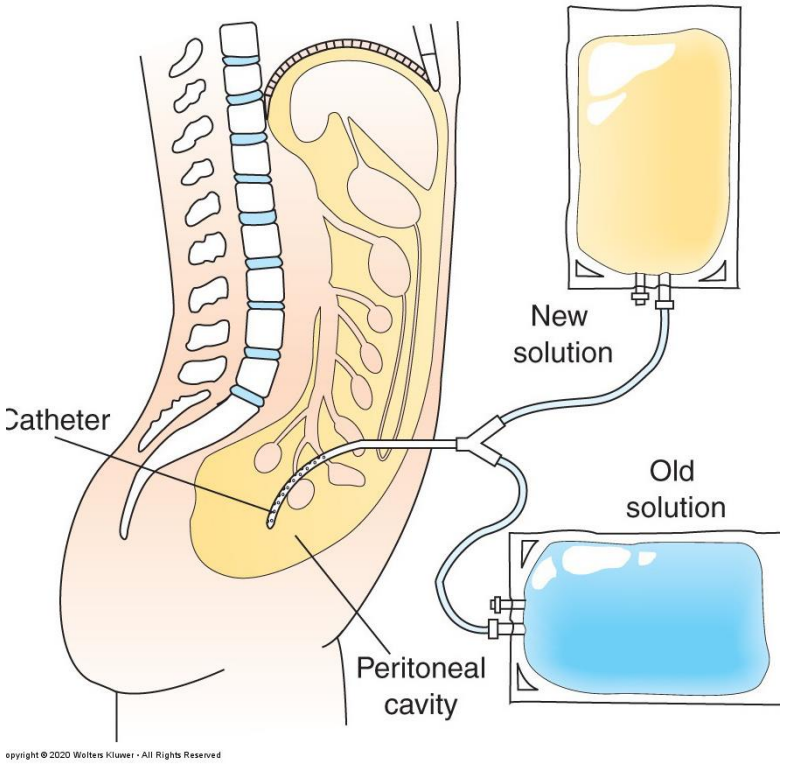
RENAL REPLACEMENT THERAPY (DIALYSIS & TRANSPLANTATION)

- **Introduction to Dialysis**
- Indications for dialysis (AEIOU: Acidosis, Electrolytes, Intoxication, Overload, Uremia)
- Types: Hemodialysis (HD) & Peritoneal Dialysis (PD)
- **Hemodialysis**
- **Mechanism:** Blood filtered through a dialyzer
- **Vascular access:** AV fistula, graft, catheter
- **Complications:** Hypotension, infections, disequilibrium syndrome



PERITONEAL DIALYSIS (PD)

- **Mechanism:** Dialysate infused into peritoneal cavity
- **Types:** Continuous Ambulatory (CAPD) vs. Automated (APD)
- **Complications:** Peritonitis, catheter infections
- **Kidney Transplantation**
- Indications and eligibility
- Living vs. cadaveric donors
- Immunosuppressive therapy
- https://youtu.be/SgBMoCArNak



Comparison & Nursing Considerations
AKI vs. CKD – Key Differences

Feature	AKI	CKD
Onset	Sudden (hours-days)	Gradual (months-years)
Reversibility	Often reversible	Irreversible
Causes	Hypovolemia, toxins	Diabetes, HTN
Management	Fluids, dialysis	Lifestyle, dialysis, transplant

NURSING INTERVENTIONS IN AKI & CKD

- Monitoring urine output and labs
- Fluid & electrolyte management
- Patient education on diet and medication adherence

- Mr. David, a 58-year-old male, presents to the outpatient nephrology clinic with complaints of persistent fatigue, swelling in his legs, and generalized itching. He also reports a loss of appetite, nausea, and occasional shortness of breath. He mentions that he has been urinating less frequently, and his urine appears foamy.
- Upon examination, the nurse notes:
- **Vital Signs:** BP 165/95 mmHg, HR 88 bpm, RR 20 breaths/min
- **Physical Exam:**
  - Bilateral pitting edema in the lower extremities
  - Pale skin with evidence of excoriation from scratching
  - Periorbital puffiness
  - Crackles on lung auscultation

- Laboratory results reveal:
- **Serum Creatinine:** 6.2 mg/dL (↑)
- **Blood Urea Nitrogen (BUN):** 75 mg/dL (↑)
- **GFR:** 18 mL/min/1.73m² (↓)
- **Hemoglobin:** 9.2 g/dL (↓)
- **Potassium:** 5.8 mEq/L (↑)
- The nephrologist diagnoses Mr. Ahmed with **Chronic Kidney Disease (CKD) Stage 4** due to diabetic nephropathy. The nurse is instructed to provide patient education on disease progression, dietary restrictions, fluid management, and preparation for possible renal replacement therapy (dialysis or transplantation).

DISCUSSION QUESTIONS:

1. What are the likely causes of Mr. David's chronic kidney disease?
2. What are the key nursing interventions for managing CKD in this patient?
3. How should the nurse educate Mr. David about dietary and fluid restrictions?
4. When should dialysis be considered in this patient, and what are the available options?
5. What are potential complications of CKD that nurses should monitor for?